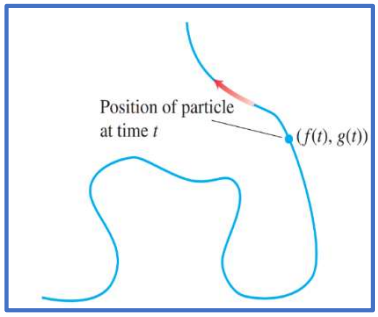
9.1 - Defining & Differentiating Parametric Equations

Learning Objective CHA-3.G: Calculate derivatives of parametric equations.

Graphing Parametric Equations



The figure to the right shows the path of a moving particle in the xy-plane. Notice that the path fails the vertical line test, so we cannot describe the graph as that of a function.
However, we can often describe the path by a pair of equations, namely $x = f(t)$ and $y = g(t)$, where f and g are continuous functions.
If this graph truly depicted the path of a moving object, say a bug walking along a glass windowpane, t would denote time.

Equations like these can describe more general curves that those described by a single function and they not only provide the graph of the path traced out but also the location of the particle (or bug) $(x, y) = (f(t), g(t))$ at any time t.

DEFINITION OF A PLANE CURVE
If f and g are continuous functions of t on an interval I, then the equations
$x = f(t)$ and $y = g(t)$
are called parametric equations and t is called the parameter. The set of points (x, y) obtained as t varies over the interval I is called the graph of the parametric equations. Taken together, the parametric equations and the graph are called a plane curve, denoted by C.

When sketching a curve by hand represented by parametric equations, you use increasing values of t. Thus, the curve will be traced over a specific **direction**. This is called the **orientation** of the curve. You use **arrows** to show the orientation.

Example 1: Sketching a Parametric Curve - I.

Sketch the curve described by the parametric equations $x = t^2 - 4$ and $y = \frac{t}{2}$, $-2 \le t \le 3$.

t	-2	-1	0	1	2	3
x	0	-3	-4	-3	0	5
y	-1	-0.5	0	0.5	1	1.5

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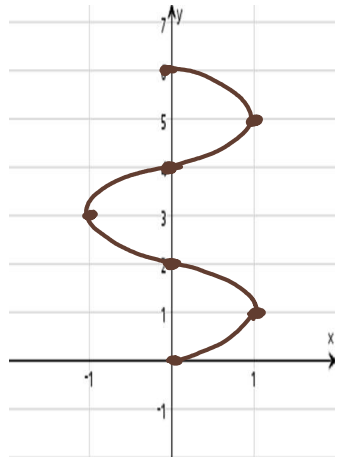
Example 2: Sketching a Parametric Curve - II

Sketch the curve described by the parametric equations

$x = \sin\left(\frac{\pi t}{2}\right)$ and $y = t$ for $0 \le t \le 6$.

t	1	2	3	4	5	6
x	1	0	−1	0	1	0
y	1	2	3	4	5	6



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Eliminating Parameters

Many times, when a parametric equation is given, we wish only to sketch the general shape of the plane curve. In that case, we wish to **eliminate the parameter** to create a rectangular equation in the form of $y = f(x)$. The technique to accomplish this is to solve for the parameter in one of the parametric equations (choosing the easiest to do so) and then replacing the result in the other equation.

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Example 3: Adjusting the Domain After Eliminating the Parameter

Sketch the curve described by the parametric equations

$x = \frac{1}{\sqrt{t+1}}$ and $y = \frac{t}{t+1}$, $t > -1$

by eliminating the parameter and adjusting the domain of resulting rectangular equation.

$\lim_{t\to\infty} \frac{1}{\sqrt{t+1}} = 0$

$\lim_{t\to\infty} \frac{t}{t+1} = 1$

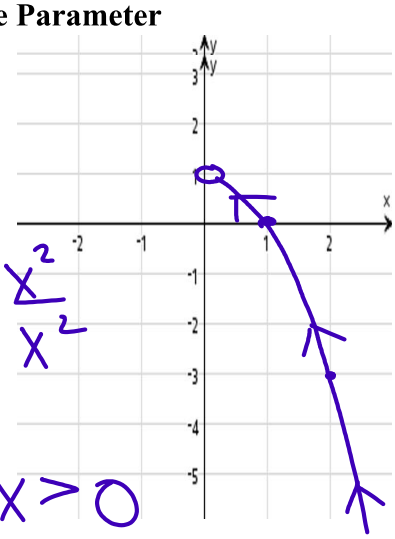
$\sqrt{t+1} = \frac{1}{x}$

$t+1 = \frac{1}{x^2}$

$t = \frac{1}{x^2} - 1$

$y = \frac{\frac{1}{x^2} - 1}{\frac{1}{x^2}} \cdot \frac{x^2}{x^2}$

$y = 1 - x^2, x > 0$



DEFINITION: SMOOTH CURVE

Let C denote a plane curve represented by the parametric equations $x = f(t)$ and $y = g(t)$ for $a \le t \le b$. Suppose each function $x(t)$ and $y(t)$ is continuous on the closed interval $[a,b]$ and differentiable on the open interval (a, b). If both $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are both continuous and are never simultaneously 0 on (a, b), then C is called a **smooth curve** on $[a,b]$.

Differentiating Parametric Equations

THEOREM 9.1 PARAMETRIC FORM OF THE DERIVATIVE

If a smooth curve C is given by the equations $x = f(t)$ and $y = g(t)$, then the slope of C at (x, y) is

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \quad \frac{dx}{dt} \neq 0.$$

HORIZONTAL & VERTICAL TANGENT LINES

At a number t where $\frac{dy}{dt} = 0, \left(\text{but } \frac{dx}{dt} \neq 0\right)$, a smooth curve c has a **horizontal tangent line**.

At a number t where $\frac{dx}{dt} = 0, \left(\text{but } \frac{dy}{dt} \neq 0\right)$, a smooth curve c has a **vertical tangent line**.

Example 4: Tangent Line to a Plane Curve

Consider the plane curve defined by the parametric equations $x(t) = 3t^2$, $y(t) = 2t$.

a.) Find an equation of the tangent line to the curve when $t = 1$.

$x'(t) = 6t$

$y'(t) = 2$

$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{2}{6t} = \frac{1}{3t}$

$x(1) = 3$

$y(1) = 2$

$\left.\frac{dy}{dx}\right|_{t=1} = \frac{1}{3}$

$L(x) = 2 + \frac{1}{3}(x-3)$

b.) Find all the points on the curve at which the tangent line is vertical.

VTL if $x'(t) = 0$ but $y'(t) \neq 0$.

$x'(t) = 0 \Rightarrow 6t = 0$

$y'(0) = 2 \neq 0 \Rightarrow t = 0$

$x(0) = 0$

$y(0) = 0$

$(0, 0)$

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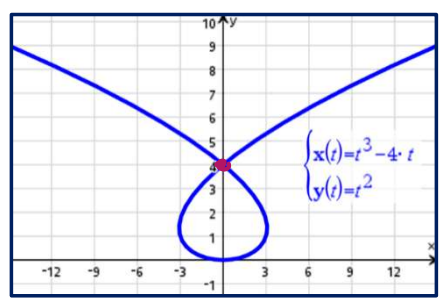
Example 5: A Curve That Crosses Itself

The plane curve represented by the parametric equations $x(t) = t^3 - 4t$, $y(t) = t^2$, crosses itself at the point (0, 4), as shown in the graph to the right, and thus has two tangent lines. Find equations for these two lines.



$x'(t) = 3t^2 - 4$

$y'(t) = 2t$

$\frac{dy}{dx} = \frac{2t}{3t^2 - 4}$

$L_1(x) = 4 + \frac{1}{2}(x - 0)$

$L_2(x) = 4 - \frac{1}{2}(x - 0)$

$t^3 - 4t = 0$

$t(t^2 - 4) = 0$

$\cancel{t = 0}$, $t = 2, t = -2$

$x(-2) = x(2) = 0$

$y(-2) = y(2) = 4$

$\left.\frac{dy}{dx}\right|_{t=-2} = \frac{-4}{8} = -\frac{1}{2}$

$\left.\frac{dy}{dx}\right|_{t=2} = \frac{1}{2}$

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<u>Example 6</u>: Projectile Motion Using Parametric Equations

A projectile is fired at an angle θ, $0<\theta<\frac{\pi}{2}$, to the horizontal with an initial speed of v_0 meters per second. Assuming no air resistance, the position of the projectile after t seconds is given by the parametric equations $x(t)=(v_0\cos\theta)t$, $y(t)=-\frac{1}{2}gt^2+(v_0\sin\theta)t$, $t\geq 0$, where g is the acceleration due to gravity.

a.) Find the slope of the tangent line to the motion of the projectile as a function of t.

$$\frac{dy}{dx}=\frac{y'(t)}{x'(t)}=\frac{-gt+v_0\sin\theta}{v_0\cos\theta}$$

$$=\frac{-gt}{v_0\cos\theta}+\tan\theta$$

$$x'(t)=v_0\cos\theta$$

$$y'(t)=-gt+v_0\sin\theta$$

b.) At what time is the projectile at its maximum height?

Max. height : $y'(t)=0$

$$\Rightarrow -gt+v_0\sin\theta=0$$

$$t_{max}=\frac{v_0}{g}\sin\theta$$

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$$y(t_{max})=-\frac{1}{2}g\left(\frac{v_0}{g}\sin\theta\right)^2+(v_0\sin\theta)\cdot\frac{v_0}{g}\sin\theta$$

$$=\boxed{\frac{v_0}{2g}\sin\theta}$$

$$x(2t_{max})=\frac{v_0^2}{g}\sin\theta\cos\theta=\frac{v_0^2}{2g}\sin 2\theta$$

$$\frac{\partial x}{\partial\theta}=\frac{v_0^2}{g}\cos 2\theta \qquad \frac{\partial x}{\partial\theta}=0 \text{ when } \theta=\frac{\pi}{4}$$

$$\therefore x_{max}=\boxed{\frac{v_0^2}{2g}}$$

9.2 - Second Derivatives of Parametric Functions

Learning Objective CHA-3.G: Calculate derivatives of parametric functions.

Because dy/dx is a function of t, you can use Theorem 9.2 repeatedly to find *higher-order* derivatives.

THEOREM 9.2 HIGHER ORDER DERIVATIVES PARAMETRICALLY

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{dx/dt} \qquad \textbf{Second derivative}$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx}\left[\frac{d^2y}{dx^2}\right] = \frac{\frac{d}{dt}\left[\frac{d^2y}{dx^2}\right]}{dx/dt} \qquad \textbf{Third derivative}$$

Example 1: Finding the Second Derivative of a Parametric Equation

Find and simplify $\frac{d^2y}{dx^2}$ for each of the following.

a.) $x(t) = e^t,\ y(t) = te^{-t}$

$x'(t) = e^t$

$y'(t) = 1\cdot e^{-t} + t\cdot e^{-t}\cdot -1$

$= e^{-t}(1-t)$

$\frac{dy}{dx} = \frac{e^{-t}(1-t)}{e^t}$

$= e^{-2t}(1-t)$

$\frac{d}{dt}\left(\frac{dy}{dx}\right) = e^{-2t}\cdot -1 - 2e^{-2t}(1-t)$

$= e^{-2t}(2t-3)$

$\therefore \frac{d^2y}{dx^2} = \frac{e^{-2t}(2t-3)}{e^t} = \boxed{\frac{2t-3}{e^{3t}}}$

b.) $x(t) = \cos t,\ y(t) = \sin 2t,\ 0 < t < \pi$

$x'(t) = -\sin t$

$y'(t) = 2\cos 2t$

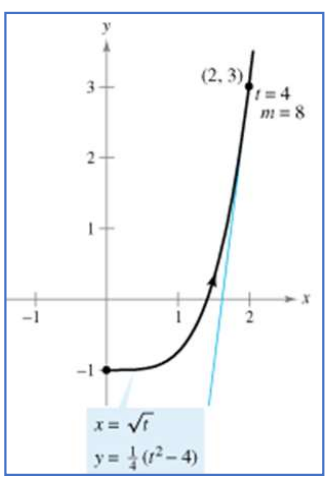
$\frac{dy}{dx} = \frac{2\cos 2t}{-\sin t} = \frac{-2\cos 2t}{\sin t}$

$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{\sin t\cdot 4\sin 2t + 2\cos 2t\cdot\cos t}{\sin^2 t}$

$= \frac{2(2\sin t\sin 2t + \cos t\cos 2t)}{\sin^2 t}$

$\therefore \frac{d^2y}{dx^2} = \boxed{\frac{-2(2\sin t\sin 2t + \cos t\cos 2t)}{\sin^3 t}}$

Example 2: Finding Slope and Concavity



For the curve given by $x = \cancel{\sqrt{t}}$ $t^{1/2}$ and $y = \frac{1}{4}(t^2 - 4),\ t \geq 0,$ find the slope and concavity at the point (2, 3).

$\frac{1}{4}(4^2 - 4) = 3$

$y = \frac{1}{4}t^2 - 1$

$\sqrt{t} = 2$

$t = 4$

$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{\frac{1}{2}t^1}{\frac{1}{2}t^{-1/2}} = t^{3/2}$

$\left.\frac{dy}{dx}\right|_{t=4} = 4^{3/2} = 8$

$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(t^{3/2})}{\frac{1}{2}t^{-1/2}} = \frac{\frac{3}{2}t^{1/2}}{\frac{1}{2}t^{-1/2}}$

$= 3t$

$\left.\frac{d^2y}{dx^2}\right|_{t=4} = 3(4) = 12$

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Fundamental Theorem of Calculus with Parametric Equations

Example 3: Position Desired

A particle moving along a curve in the xy-plane is at position $(x(t), y(t))$ at time t, where $\frac{dx}{dt} = \ln(t+1)$, $\frac{dy}{dt} = \arcsin\left(e^{-t^2}\right)$ for $t \geq 0$. At time $t = 1$ the particle is at position (2, 5).

a.) Find the slope of the tangent line to the curve at position (2, 5).

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\arcsin(e^{-t^2})}{\ln(t+1)}$$

$$\left.\frac{dy}{dx}\right|_{t=1} = \frac{\arcsin\left(\frac{1}{e}\right)}{\ln(2)} = 0.544$$

b.) Find the x-coordinate of the position of the particle at time $t = 3$.

$$\int_1^3 x'(t)\,dt = x(3) - x(1)$$

$$x(3) = 2 + \int_1^3 \ln(t+1)\,dt$$

$$x(3) = +2.159$$

$$= 4.159$$

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9.3 - Finding Arc Lengths of Curves Given by Parametric Equations

Learning Objective CHA-6.B: Determine the length of a curve in the plane defined by parametric functions, using a definite integral.

Now that we have seen how parametric equations can be used to describe the path of a particle moving in the plane, our next step is to develop a formula for determining the *distance* traveled by the particle along its path.

Recall our arc length formula for a curve defined by *y(x)* on the interval $[x_0, x_1]$.

$$L = \int_{x_0}^{x_1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

If our curve is represented by the parametric equations,
$x = f(t)$ and $y = g(t)$, $a \le t \le b$, and $dx/dy = f'(t) > 0$, you can write

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_a^b \sqrt{1 + \left(\frac{dy/dt}{dx/dt}\right)^2}\, dx = \int_a^b \sqrt{\frac{(dx/dt)^2 + (dy/dt)^2}{(dx/dt)^2}} \cdot \frac{dx}{dt}\, dt$$

$$= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt$$

THEOREM 9.3-A ARC LENGTH IN PARAMETRIC FORM

If a smooth curve C is given by the equations $x = f(t)$ and $y = g(t)$, such that C does not intersect itself on the interval $a \le t \le b$ (except possible at the endpoints), then the arc length of C over the interval is given by

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt$$

NOTE: When applying the arc length formula to a curve, be sure the curve is traced out only once on the interval of integration. For instance, $x = \cos t$ and $y = \sin t$ is traced out once on the interval $0 \le t \le 2\pi$, but is traced out twice on the interval $0 \le t \le 4\pi$.

ARC LENGTH & DISTANCE TRAVELED

The distance a particle travels along a smooth curve on the interval $a \le t \le b$ is synonymous with the arc length of the curve over that same interval. Or,

$$\text{Distance} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt$$

Example 1: Finding Arc Length for a Parametrically Defined Curve

Find the arc length of the curve defined by $x(t) = t^3 + 2,\ y(t) = 2t^{9/2}$ on the interval $1 \le t \le 3$.

$x'(t) = 3t^2$

$y'(t) = 2 \cdot \frac{9}{2} t^{7/2} = 9t^{7/2}$

$L = \int_1^3 \sqrt{(3t^2)^2 + (9t^{7/2})^2}\, dt$

$L = \int_1^3 \sqrt{9t^4 + 81t^7}\, dt$

$u = 1 + 9t^3$

$du = 27t^2 dt$

$L = \int_1^3 3t^2 \sqrt{1 + 9t^3}\, dt$

$L = \frac{1}{9} \int_{10}^{244} u^{1/2}\, du$

$L = \left[\frac{1}{9} \cdot \frac{2}{3} u^{3/2}\right]_{10}^{244} = \frac{2}{27}\left(244^{3/2} - 10^{3/2}\right)$

$= 279.984$

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Example 2: Finding Distance Traveled

A particle moving along a curve in the xy-plane is at position $(x(t), y(t))$ at time t, for $t \ge 0$, where $\frac{dx}{dt} = \tan^{-1}(e^{-t})$, $\frac{dy}{dt} = \frac{3t}{1+4t^3}$. Find the distance traveled by the particle on the time interval $1 \le t \le 2$.

$L = \int_1^2 \sqrt{\left(\tan^{-1}(e^{-t})\right)^2 + \left(\frac{3t}{1+4t^3}\right)^2}\, dt$

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